`scripts/run-challenge-matrix.sh` — User Guide

Last verified: 2026-05-16 (HelixQA Option 1 wiring landed — `--include-helixqa` flag added) **Inheritance:** HelixConstitution §11.4.18 + Lava §6.AE.6 + §6.X (Container-Submodule Emulator Wiring) + §6.I (Multi-Emulator Container Matrix)

Overview

Operator entry point for §6.AE gate-mode Challenge Test runs. Pre-bakes the §6.AE.2 minimum AVD matrix (API 28 / 30 / 34 / latest stable × phone + tablet) and delegates to `submodules/containers/cmd/emulator-matrix --runner=containerized` per §6.X.

Honest pre-flight

The script DETECTS the §6.X-debt darwin/arm64 host gap (no `/dev/kvm` available to podman containers; macOS HVF not exposed to podman) and, when detected, REFUSES to claim a successful gate run. Instead it:

1. Validates arguments + matrix minimum
2. Writes `<evidence-dir>/host-preflight.json` with the host-gap classification
3. Exits 2 (NOT 0) — gate-host ineligible

Per §6.J/§6.L: no false-pass; honest unblock report. Real gate runs require a Linux x86_64 + KVM host.

Usage

```bash
# Run ALL Challenges on the §6.AE.2 minimum matrix
bash scripts/run-challenge-matrix.sh

# Run a specific Challenge
bash scripts/run-challenge-matrix.sh \
  --test-class
       lava.app.challenges.Challenge01AppLaunchAndTrackerSelectionTest

# Add TV-class AVD (when feature touches TvActivity)
bash scripts/run-challenge-matrix.sh --add-tv
```

```bash
# Add foldable AVD
bash scripts/run-challenge-matrix.sh --add-foldable

# Override "latest stable" API level
bash scripts/run-challenge-matrix.sh --latest-api 36

# Skip APK rebuild
bash scripts/run-challenge-matrix.sh --no-build

# Override the matrix entirely with explicit AVDs (target EXISTING
        host AVDs).
# Use on a macOS host-direct+HVF run where the default CZ_API*
        images are not provisioned.
bash scripts/run-challenge-matrix.sh --avds
        "CZ_API35_Phone_Fresh:35:phone" --no-build

# ALSO invoke the 11 HelixQA Challenge scripts (Option 1 wiring)
bash scripts/run-challenge-matrix.sh --include-helixqa
```

Inputs

Arg	Description
`--test-class <fqn>`	Specific Challenge to run (default: all under `lava.app.challenges`)
`--evidence-dir <dir>`	Output directory (default: dated under `.lava-ci-evidence/`)
`--no-build`	Skip the `:app:assembleDebug` step
`--avds "<name:api:form[,...]>"`	**Replace** the §6.AE.2 default matrix entirely with an explicit comma-separated AVD list. Use to target **existing host AVDs** (e.g. `CZ_API35_Phone_Fresh:35:phone` on a macOS host-direct+HVF run where the default `CZ_API*` images are not provisioned). When set, `--latest-api` / `--add-tv` / `--add-foldable` are ignored. A sub-minimum list is a **development-iteration** run, **not** a §6.AE.2-conformant gate matrix. NOTE: the Containers CLI still resolves each AVD's system image through `tools/lava-containers/vm-images.json` (`--image-manifest`); the manifest must contain a matching image id (e.g. `android-35-phone`) or the runner fails with `no image with id`.
`--latest-api <N>`	Override the "latest stable" API level (default: 36; ignored when `--avds` is set)
`--add-tv`	Add a TV-class AVD to the matrix
`--add-foldable`	Add a foldable AVD
`--include-helixqa`	ALSO invoke `scripts/run-helixqa-challenges.sh` (the 11 HelixQA Challenge scripts per Option 1 wiring). OFF by default so existing matrix runs are unaffected. HelixQA runs

Arg	Description
	on the HOST BEFORE the AVD matrix → independent of §6.X-debt darwin/arm64 gate-host gap. HelixQA wrapper evidence lands at `<evidence-dir>/helixqa/`. Non-zero HelixQA exit promotes the final aggregate exit code (matrix exit dominates if both fail). See `docs/scripts/run-helixqa-challenges.sh.md` for full details.

§6.AE.2 mandatory minimum matrix

```
CZ_API28_Phone:28:phone
CZ_API30_Phone:30:phone
CZ_API34_Phone:34:phone
CZ_API34_Tablet:34:tablet
CZ_API<latest>_Phone:<latest>:phone
```

Plus TV / foldable when explicitly requested by `--add-tv` / `--add-foldable`.

Sub-minimums are permitted for development iteration; the gate row's `gating: true` flag is only set when the full minimum is satisfied + every config dimension (theme/locale/density per §6.AE.2) is covered.

Outputs

- `<evidence-dir>/host-preflight.json` — host-gap classification (always written)
- `<evidence-dir>/real-device-verification.{md,json}` — per-AVD attestation rows (only on Linux x86_64 + KVM gate-host)
- `<evidence-dir>/helixqa/helixqa-attestation.json` + per-script logs — only when `--include-helixqa` is set

Exit codes

Code	Meaning
0	Matrix completed; all AVDs passed
1	At least one AVD failed boot OR at least one test failed (real Containers CLI exit)
2	Gate-host ineligible (darwin/arm64 OR no KVM); honest unblock-needed report written

Cross-references

- `submodules/containers/cmd/emulator-matrix` (the underlying runner)
- `tools/lava-containers/vm-images.json` (matrix manifest)

- `scripts/run-emulator-tests.sh` (older sister-glue; same delegation, different default arguments)
- `scripts/run-helixqa-challenges.sh` + `docs/scripts/run-helixqa-challenges.sh.md` (the HelixQA wrapper invoked by `--include-helixqa`)
- `docs/plans/2026-05-16-helixqa-integration-design.md` (Option 1 wiring design)
- `.lava-ci-evidence/sixth-law-incidents/2026-05-13-emulator-container-darwin-arm64-gap.json` (the standing §6.X-debt incident)
- Lava `CLAUDE.md` §6.AE + §6.X + §6.I